

RFQ Analysis

The DermaPod™ camera hardware subsystem requirements specified in the RFQ (AIS-DPD-2026-001 Rev B) are divided into five primary engineering sections:

- 1. Camera Geometry & Architecture
- 2. Image Calibration & Quality
- 3. Mechanical & Pose Repeatability of Robot
- 4. Illumination Subsystem
- 5. Memory & Data Management

Primary Architectural Goal & System Complexity

The most complex engineering challenge when defining the system architecture is simultaneously satisfying four coupled primary parameters:

- Resolving Power ROI: 30 px/mm baseline spatial resolution across the target skin surface (IMG-3).
- Depth of Field (DoF ≥ 100 mm): Maintaining focus across full body curvature (IMG-7).
- Patient Cycle Time: Total session execution time ≤ 90 seconds (ROB-5).
- Full Envelope & 98% Body Coverage: Achieving 98% anatomical skin coverage across the 1.0 m (W) $\times$ 0.6 m (D) $\times$ 2.0 m (H) patient scan volume (MEC-1 / IMG-5).

Circle of Confusion (CoC) Definition

Definition & Assumption: Before evaluating optical Depth of Field (DoF), we must first define the Circle of Confusion (CoC). Because CoC is not explicitly defined in the RFQ document for the DoF requirement (IMG-7), we have established $\text{CoC} = 25 \mu\text{m}$ based on sensor pixel pitch and target spatial resolving power.

Optical Trade-Off & System Architecture Analysis

- Pixel Density vs. FOV: As required pixel density increases (to achieve 30 px/mm), single-camera Field of View (FOV) decreases proportionally.
- FOV vs. Depth of Field (DoF): As Field of View (FOV) decreases (higher magnification), optical Depth of Field (DoF) also decreases. Maintaining $\text{DoF} \geq 100$ mm across smaller FOVs requires smaller optical apertures (higher f-number), which in turn demands higher illumination power.
- High-Resolution Sensor Strategy & Cost Trade-Off: By selecting high-resolution sensors (≥ 60 MP to 105 MP), each camera captures a significantly larger FOV while preserving 30 px/mm spatial density. Although individual high-resolution camera unit cost increases, it prevents an exponential increase in total camera count, thereby optimizing overall subsystem BOM cost, cabling, and optical DoF.

Proposed Architectural Models

We proposed two distinct architectural models for evaluation:

- Model 1 (2D Dual-Column Model): A dual-column array optimized for fast multi-pose 2D clinical photography with minimal gantry motion steps.
- Model 2 (2D Scanning Carriage Model): A vertical carriage scanning gantry combined with a rotating patient turntable for controlled 2D surface imaging.

Benchmark Comparison (iTBoS vs. Commercial Systems)

- iTBoS Reference: Engineered specifically for uniform pixel density across skin contours via complex 3D calibration.
- Commercial Reality (Vectra, FotoFinder, IntelliStudio): All commercial systems have variable pixel density across body contours due to distance variations.
- DermaPod Alignment: Aligns with commercial standards, ensuring worst-case distance still satisfies baseline 30 px/mm ROI.

Reference Parameters Used in Calculations:

- Lens Aperture: f/8
- Body Curvature Target: 100 mm
- Mechanical Settle Time: 2.0 seconds per step
- Turntable & Column Rotation Time: 5.0 seconds per transit
- Focus Slice Exposure Time: 0.2 seconds

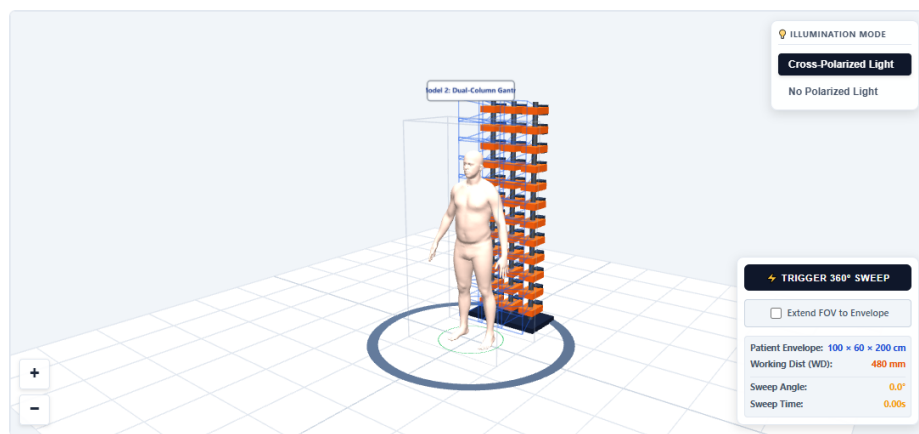
2. 3D Simulation & Architectural Evaluation

Model 1: Option A — Dual-Column Parallel Gantry

- Mechanical: Multiple parallel columns (upgraded to 3 columns to span the horizontal 1000mm width at 30 px/mm resolving power) on an elliptical track.
- Acquisition: Performs a segmented step-by-step horizontal rotation sweep, stopping at 4 discrete poses (0°, 90°, 180°, 270°) to capture complete patient coverage.
- Cost: High camera count and high BOM cost.
- Risk: Calibration complexity for multi-column geometries; high data ingestion bandwidth.
- Timing Considerations: Scan time is dictated by the column rotation transit and stop exposure time. Baselines assume 5.0s rotation transit time and 2.0s settle time per step.

Model 1 Sensor Optimization Matrix (Target: 30 px/mm, 3 Columns, CoC = 25 μ m, Aperture = f/8):

Sensor Profile	FOV (W×H)	Total Array FOV	Horiz Steps	Vert Steps	Combined Steps	Avg Pixel Density	Min Pixel Density	DoF (CoC=2 5µm)	Cams Needed	Total Cost	Scan Time
GMAX3265 (65.4MP)	311mm × 233mm	840mm × 2176mm	4 steps	1 step	4 steps	31.5 px/mm	24.7 px/mm	82.8 mm	33 (3 × 11)	\$214,500	26.2s (+20s to +30s)
IMX927 / IMX937 (105MP)	341mm × 341mm	921mm × 2028mm	4 steps	1 step	4 steps	31.3 px/mm	24.9 px/mm	98.0 mm	21 (3 × 7)	\$178,500	26.2s (+20s to +30s)
Sony IMX661 (127.7MP)	445mm × 317mm	1201mm × 2159mm	4 steps	1 step	4 steps	32.2 px/mm	23.8 px/mm	41.1 mm	24 (3 × 8)	\$288,000	27.8s (+20s to +30s)
Sony IMX411 (151MP)	473mm × 354mm	1277mm × 2106mm	4 steps	1 step	4 steps	32.6 px/mm	23.4 px/mm	31.4 mm	21 (3 × 7)	\$315,000	26.2s (+20s to +30s)

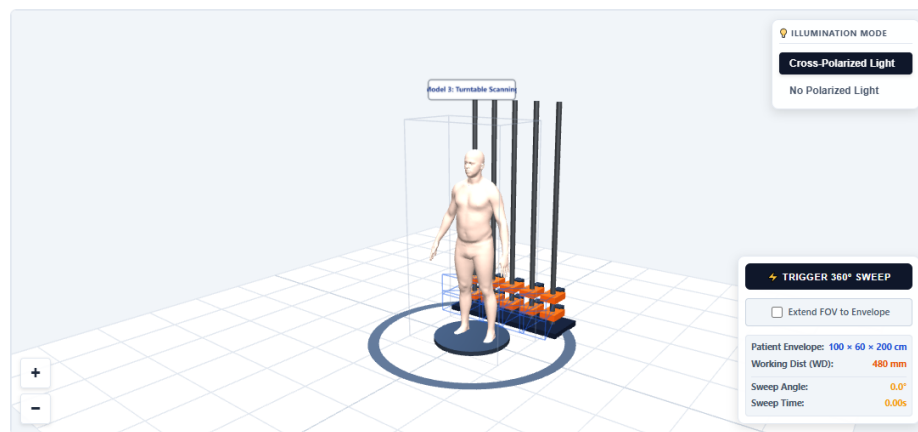


Model 2: Option B — Turntable Scanning Gantry

- Mechanical: Static columns with motorized vertical carriages; patient stands on a rotating turntable.
- Acquisition: Turntable rotates patient to 4 poses (0°, 90°, 180°, 270°); carriages sweep vertically at each stop.
- Cost: Low camera count, low BOM cost.
- Patient Pose Guidelines: Patient must dynamically adjust leg stance (forward/backward) and stretch hands based on the active camera pose location to prevent occlusions.
- Timing Considerations: Scan time is dependent on turntable rotation speed and vertical carriage travel steps. Baselines assume 5.0s turntable rotation time and 2.0s settle time per vertical sweep step.

Model 2 Sensor Optimization Matrix (Target: 30 px/mm, CoC = 25 μ m, Aperture = f/8, Carriage Travel):

Sensor Profile	FOV (W×H)	Total Array FOV	Horiz Steps	Vert Steps	Combined Steps	Avg Pixel Density	Min Pixel Density	DoF (CoC=25 μ m)	Cams Needed	Total Cost	Scan Time
GMAX3265 (65.4MP)	311mm × 233mm	1057mm × 594mm	4 steps	9 steps	36 steps	31.5 px/mm	24.7 px/mm	82.8 mm	12 (4 × 3)	\$78,000	115.8s (+20s to +30s)
IMX927 / IMX937 (105MP)	341mm × 341mm	1159mm × 870mm	4 steps	5 steps	20 steps	31.3 px/mm	24.9 px/mm	98.0 mm	9 (3 × 3)	\$76,500	71.4s (+20s to +30s)
Sony IMX661 (127.7MP)	445mm × 317mm	1133mm × 807mm	4 steps	6 steps	24 steps	32.2 px/mm	23.8 px/mm	41.1 mm	9 (3 × 3)	\$108,000	91.8s (+20s to +30s)
Sony IMX411 (151MP)	473mm × 354mm	1206mm × 903mm	4 steps	5 steps	20 steps	32.6 px/mm	23.4 px/mm	31.4 mm	9 (3 × 3)	\$135,000	71.4s (+20s to +30s)



Important Notes & Pricing Disclaimers:

- **Budgetary Estimates Only:** All camera unit costs and total system estimates listed in this document are budgetary catalog figures and subject to change once formal firm-fixed quotes are received directly from camera manufacturers and authorized distributors.
- **Scope of Pricing (Camera Bodies Only):** The pricing listed reflects camera bodies only. Costs do NOT include optical lenses, lens mounts, frame grabbers, interface cabling, illumination arrays, power supplies, or mechanical gantry structures.

Architectural Classification: 3D Reconstruction vs. 2D Separate Image Analysis

- Model 1 & Model 2 (2D Clinical Photography Systems): Model 1 (Dual-Column) and Model 2 (Scanning Carriage) are 2D Clinical Photography Systems. They execute discrete multi-pose indexing steps (0°, 90°, 180°, 270°).
- Separate Frame-by-Frame Image Analysis: In Model 1 and Model 2, diagnostic image analysis, lesion segmentation, and color profiling are performed on each captured 2D image separately (frame-by-frame analysis), rather than generating a fused 3D volumetric mesh.

Model-Specific Engineering Considerations, Motion Artifacts & Scan Time Trade-Offs

- Model 1 — Optimal 4-Pose Positioning: Model 1 relies on determining the optimal standardized 4-pose clinical positions (0°, 90°, 180°, 270°) to maximize anatomical line-of-sight and achieve 98% skin surface coverage while keeping patient repositioning simple.
- Model 2 — 4-Pose Optimization & Extended Scan Time: Like Model 1, Model 2 uses standardized 4-pose positions (0°, 90°, 180°, 270°). However, Model 2 takes significantly more total scan time (71.4s – 115.8s depending on sensor FOV) compared to Model 1 (26.2s + 20s to 30s overhead) because it requires combined turntable rotation transits and multi-step vertical carriage travel.
- Estimated Scan Timing & Optical Scope Note: All scan timing figures listed across Model 1 and Model 2 are rough estimates based on baseline transit assumptions. They do NOT account for exact motor acceleration/deceleration profiles, jerk-limited kinematics, or firmware handshake latencies. The primary objective of this phase is strictly optical and geometric modeling — specifically calculating Pixel Density (px/mm), Field of View (FOV W×H), Camera Array Count, and Depth of Field (DoF) to achieve 98% patient envelope coverage.

Interactive 3D Simulation & Web Calculator Access:

- Live Cloudflare Deployment URL: <https://poc6-4pc.pages.dev/>
- Security Passkey / Password: rfq@30
- Simulation Features: Provides real-time interactive 3D WebGL gantry manipulation, patient envelope positioning, real-time optical resolution calculations, camera array FOV coverage, and multi-user collaborative RFQ feedback.

Optical Physics Formulas & Authoritative Technical Resource Links

Consolidated optical physics formulas, spatial resolution calculations, and Depth of Field equations verified against standard optics references:

- **1. Optical Magnification (M) & Field of View (FOV):**
 Formulas: $M = f / (WD - f)$, $FOV_W = W_sensor / M$, $FOV_H = H_sensor / M$.
 (where f = focal length, WD = Working Distance, W_sensor / H_sensor = active sensor dimensions in mm).
 Reference Resource: Edmund Optics — Understanding Focal Length and Field of View
 (<https://www.edmundoptics.com/knowledge-center/application-notes/imaging/understanding-focal-length-and-field-of-view/>)

- **2. Spatial Resolving Power & Pixel Density Calculation:**

Formulas: Pixel Density (px/mm) = $W_{\text{pixels}} / \text{FOV}_W = H_{\text{pixels}} / \text{FOV}_H = M / \text{Pixel_Pitch_mm}$.

(where Pixel_Pitch_mm = pixel pitch in $\mu\text{m} / 1000$, target baseline = 30.0 px/mm).

Reference Resource: Edmund Optics — Resolution and Pixel Size in Imaging Systems

(<https://www.edmundoptics.com/knowledge-center/application-notes/imaging/resolution/>)

- Formulas: Hyperfocal $H = (f^2 / (N * c)) + f$, $D_{\text{near}} = (s * f^2) / (f^2 + N * c * (s - f))$, $D_{\text{far}} = (s * f^2) / (f^2 - N * c * (s - f))$,

$\text{DoF} = D_{\text{far}} - D_{\text{near}}$, Focus Stacking Slices = $\text{ceil}(\max(0, \text{Target Body Depth (100 mm)} - \text{DoF}) / (\text{DoF} * (1 - \text{Stack_Overlap}))) + 1$.

- **4. Diffraction Limit & Airy Disk Diameter:**

Formula: $d_{\text{Airy}} = 2.44 * \text{wavelength} * N$ (where wavelength = 0.55 μm green light, $N = f/8$ aperture).

Reference Resource: Edmund Optics — The Airy Disk and Diffraction Limit

(<https://www.edmundoptics.com/knowledge-center/application-notes/imaging/limitations-on-resolution-the-airy-disk/>)

- **5. Angular Field of View (AFOV):**

Formulas: $\text{AFOV}_H = 2 * \arctan(W_{\text{sensor}} / (2 * f))$, $\text{AFOV}_V = 2 * \arctan(H_{\text{sensor}} / (2 * f))$.

Reference Resource: Edmund Optics — Calculating Angular Field of View

(<https://www.edmundoptics.com/knowledge-center/application-notes/imaging/understanding-focal-length-and-field-of-view/>)